Grouping in pairs

1 Factorise:

a $5(a+b)+v(a+b)$

b $x(y-z)-w(y-z)$

c $5y(4w+3x)-z(4w+3x)$

d $2y\left(2x^2+3z\right)-\left(2x^2+3z\right)$

e $3f(g+h)+(g+h)^2$

f $8y(y-4)+3(4-y)$

g $(2c-d)(c+5d)-3(d-2c)$

h $3x^2(x+4y)-5y(x+4y)$

2 Factorise:

a $8x+xz-16y-2yz$

b $24+3y+8x+xy$

c $7xy+wx+7yz+wz$

d $x^2+2x+5x+10$

e $x^2-3x+8x-24$

f $2mp+6+3p+4m$

g a^3+8a^2+a+8

h $3pq^2-11ypq+3rsq-11yrs$

i $4x+24yz+32xy+3z$

j $2x+18yz+12xy+3z$

Perfect squares

3 Factorise:

a $q^2+2qt+t^2$

b $u^2-2uq+q^2$

c $b^2-2br+r^2$

d x^2+6x+9

e c^2-4c+4

f $64+2e+e^2$

g $49-2p+p^2$

h $16w^2-40w+25$

i $64r^2+48r+9$

j $\frac{1}{4}-3d+9d^2$

4 Factorise:

a x^4+8x^2+16

b a^4-18a^2+81

5 A square has area $t^2+14t+49$.

a Find the side length of the square.

b Find the perimeter of the square.

Difference of two squares

6 One linear factor of $x^2 - 16$ is $x - 4$. Find the other factor.

7 What can we multiply $z + 7$ by to give the difference of two squares?

8 Factorise:

a $x^2 - y^2$
b $n^2 - 25$
c $v^2 - 1$
d $121 - v^2$
e $x^2 - \frac{1}{4}$
f $x^2 - \frac{25}{121}$
g $16 - 9y^2$
h $x^2y^2 - 49$
i $25m^2 - 49$
j $81x^2 - 16y^2$
k $7x^2 - 63$
l $45t^2 - 20$

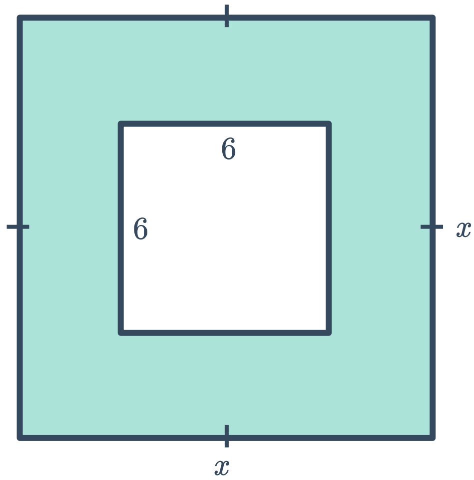
9 Using the difference of two squares, evaluate the following:

a $65^2 - 63^2$
b $99^2 - 97^2$
c $45^2 - 42^2$
d $58^2 - 52^2$

10 Find the value of the following by rewriting in the form $(x + y)(x - y)$:

a 504×49
b 304×296

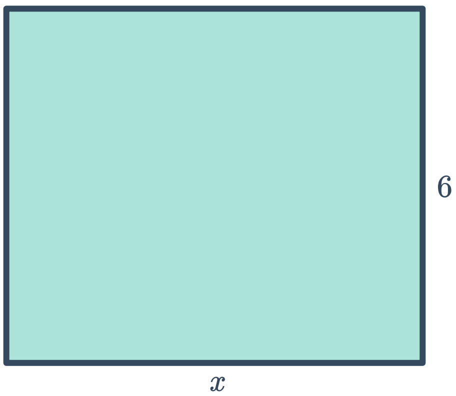
11 Consider the shaded area shown:



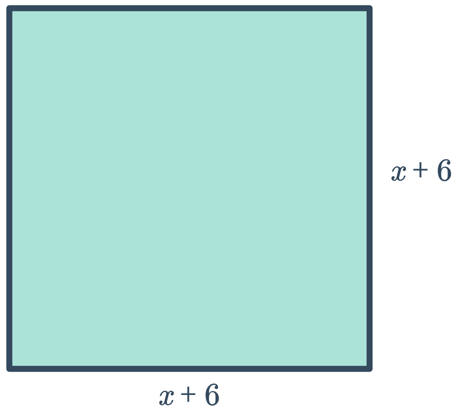
a Find the area of the shaded region.

b State whether the following rectangles have the same area as the shaded region.

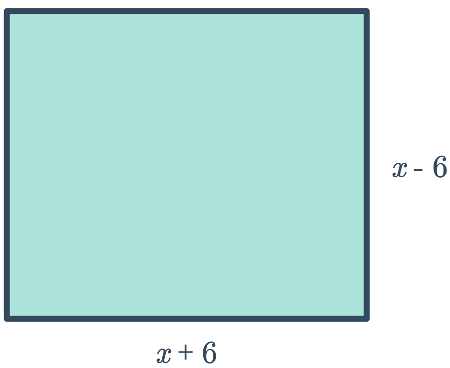
i



ii



iii



Mixed factorisations

12 Factorise:

a $3x^2y - 6y^4 + 9x^2y^3$

b $16x^2y^2 + 24xyz + 9z^2$

c $121c^2 - 49d^2$

d $3x^2y + x^2z + 3y^2 + yz$

e $a^2 + 2ab + b^2 - c^2$

f $8a^2b^4 + 12bc^2$

g $(b+4)^2 - (3-b)^2$

h $14h^3p^2r + 21h^5p^2 + 49hp^4$

i $8w^2 - 32wt^2 + 3wt - 12t^3$

j $16q^4 + 24q^2r + 9r^2$

k $48df^2 - 3dh^2$

l $128t^3 - 18s^4t$

m $18j^2 + 60jk + 50k^2$

n $x^2y + 7x^2 - y - 7$

13 Factorise $x^4 - 1$.